Ahtsham Mustafa Awan

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Research Interests & SKILLS

- **Research:** Extreme Events Risk, Compound Hazards, Geospatial Artificial Intelligence, Machine/Deep Learning, Spatial Hydrology, Web GIS, Spatial Data Science
- **Programming & Databases:** Python, R, C, JavaScript, SQL, ASP.NET, PostgreSQL/PostGIS
- **Software:** QGIS, ArcGIS Pro, ERDAS Imagine, ENVI, GeoServer, Global Mapper, eCognition, GeoDa
- **Hydrological / Hazard Modeling:** HEC-RAS, RAMMS, SCS-CN/AMC modeling, wflow, SWAT, geomorphometric flood-risk ranking, pysheds terrain analysis
- **Geotechnical Software:** Slide2, Plaxis 2D/3D, GeoStudio (SLOPE/W, SEEP/W), RocFall, Dips, Swedge, RS2, RockWorks

EDUCATION

MASTER OF SCIENCE IN REMOTE SENSING AND GEOGRAPHICAL INFORMATION SYSTEMS **(Sep 2021 - Jul 2024)**
National University of Sciences and Technology (NUST) ***Islamabad, Pakistan***

- **Final grade:** 3.55/4.0
- **Thesis:** Comparison of different Machine Learning Algorithms for Land Use Land Cover Change Detection and its Relationship with Land Surface Temperature and Topography
- **Coursework:** Spatial Analysis and Modeling; Advanced Remote Sensing and Digital Image Processing; Advanced Geodatabase and Programming; Natural Hazards and Disaster Management

BACHELOR OF SCIENCE (B.S.) IN APPLIED GEOLOGY **(Oct 2016 - Feb 2021)**
The University of Azad Jammu and Kashmir (UAJK) ***Muzaffarabad, Pakistan***

- **Final grade:** 3.54/4.0
- **Thesis:** Landslide Mapping and Spatial Distribution Analysis from Muzaffarabad to Luat, Neelum Valley, Azad Kashmir, Pakistan
- **Coursework:** Geomorphology, Structural Geology, Engineering Geology, Field Geology, Surveying and Drawing, Remote Sensing and GIS, Geomechanics, Geohydrology, Environmental Hazards, Geotechnical Engineering, Mathematics (I, II), Physics (I, II)

EXPERIENCE

Assistant Research Officer **(Jan 2024 - Current)**
Government of Pakistan - Ministry of Defense ***Islamabad, PK***

- Numerical modeling for disaster management, photogrammetry, slope mapping, terrain analysis, and infrastructure mapping using diverse datasets, including SPOT, Pleiades, Superview, PRSS, SAR, and LIDAR
- Conducted drone and land surveys for advanced hydrological modelling and post landslide and flood impact assessment
- Led in the development of state of the art web applications (Flood Monitoring, Spatial Data Analytics, Object Detection and AI Agents Dashboards) for geospatial data analysis and visualization

Research Assistant **(Jan 2023 - Dec 2023)**
National University of Sciences and Technology ***Islamabad, PK***

- Mapped extreme weather events (floods, landslides, and heatwaves) using remote sensing and artificial intelligence techniques
- Performed numerical modeling and simulations for coastal inundation vulnerability and flood hazard evaluations

Remote Sensing and GIS Intern **(Jan 2021 - Aug 2021)**
Pakistan Council of Research in Water Resources ***Islamabad, PK***

- Identification of groundwater potential zones using GIS/remote sensing techniques and AHP
- Conducted water quality testing, surveying, and mapping for environmental assessments
- Prepared comprehensive reports and documentation based on survey findings and analysis

PROJECTS & RESEARCH

- **SAR-Optical Flood Segmentation Framework - 2025 Pakistan floods:** a pixel-level flood segmentation architecture combining U-Net decoders with ResNet-50, DenseNet-121, and EfficientNet-B0 encoders, fusing Sentinel-1 SAR, Sentinel-2 optical imagery, and DEM-derived hydro-geomorphological indices (slope, TWI, HAND); terrain preprocessing is built around pysheds and aligned to the Sen1Floods11 benchmark
- **Multi-Sensor Flood Scene Classification (SEN12-FLOOD) - 2022 Pakistan floods:** fine-tuned four deep learning backbones (ResNet-50, DenseNet-121, EfficientNet-B0, ViT) on fused SAR/optical imagery for flood scene classification, then combined them through soft-voting and stacking ensembles (logistic regression, XGBoost, SVM), reaching 95.6% test accuracy
- **Landslide Susceptibility Mapping, Azad Kashmir:** utilised GeoAI and deep learning architectures (U-Net, ResNet, ConvLSTM) alongside spatial multicriteria decision analysis (AHP and Fuzzy Logic) to analyze geotechnical testing data (Atterberg, Casagrande, direct shear, triaxial, sieve) and multi-source Earth observation data (Sentinel, Maxar, Planet Labs, Pleiades); executed repeat-pass InSAR time-series analysis to identify micro-topography deformation, trace slope acceleration boundaries, and quantify active landslide displacements across high-risk corridors
- **Mangla Watershed Runoff & AMC Modeling:** developed a Google Earth Engine algorithm fusing MODIS LULC and USDA soil data to automate AMC maps and SCS-CN hydrological modeling, analyzing 2005-2015 runoff for the Mangla watershed; validated CHIRPS, GPM, and TRMM outputs against ground metrics, identifying CHIRPS as achieving up to 90% accuracy
- **Jhelum River Basin Flood Susceptibility Mapping:** modelled flood susceptibility using ASTER GDEM data in ArcMap with an optimized geomorphometric ranking approach, evaluating how basin gradient, relief, bedrock geology, and sub-basin area drive flash and riverine flood risks to inform government risk reduction strategies

- **Pakistan Flood Monitoring Dashboard:** led development of a full-stack dashboard using React, Leaflet.js, and Node.js to ingest real-time WAPDA telemetry and automate Sentinel-1 SAR raster processing; architectured scalable PostgreSQL/PostGIS databases and GeoServer pipelines, reducing geospatial query latency while streaming live river discharge analytics and emergency alerts
- **Badswat Glacier Debris Flow Hazard Assessment:** modelled and simulated debris flow hazard using the RAMMS model and HEC-RAS
- **Water Quality Modeling Using Landsat 8 & Hyperspectral Data:** modeled water quality parameters using Landsat 8 and ASD hyperspectral spectroradiometer data
- **Coastal Inundation Vulnerability, Indus River Delta:** simulated coastal inundation vulnerability and impact assessment using HEC-RAS
- **Himalayan Landscape Dynamics: LULC & LST Analysis (2002-2026):** analyzed twenty four years of Himalayan landscape dynamics by deploying SVM classification on Landsat imagery to map LULC changes against split-window algorithm retrieved land surface temperatures; modeled strong LST-NDVI correlations (R^2 up to 0.87) to quantify glacier decline (6.37%) and barren expansion (6.47%) for climate resilience planning
- **Kohat City Urban LULC & LST Analysis (1998-2018):** analyzed LULC dynamics and land surface temperature using multi-temporal Landsat imagery, quantifying a 55% built-up expansion and an increase in mean LST and Urban Heat Islands, demonstrating the direct impact of urban growth on microclimate changes
- **Peshawar Groundwater Quality Assessment:** evaluated the impact of urbanization and agricultural land-use changes on groundwater by classifying multi-temporal Landsat images and analyzing 217 tube well samples across 7 parameters; correlated anthropogenic pollution with declining water quality and rising health risks, documenting a 14% increase in methemoglobinemia alongside elevated cases of UTIs and peptic ulcers
- **Aedes Albopictus Distribution Modeling, Rawalpindi:** constructed high-resolution bioclimatic and LULC datasets from satellite data to model Aedes albopictus distribution using the MaxEnt algorithm; identified roadside green patches, winter temperatures, and household waste practices as primary risk factors, incorporating community surveys to guide targeted vector control

FIELDWORK

- Conducted extensive field mapping and geotechnical investigations across the Salt Range (eastern, central, and western regions) for structural analysis, stratigraphic profiling, and slope stability assessment
- Performed flood impact assessment and debris flow hazard analysis in Northern Pakistan, including Azad Kashmir, Gilgit-Baltistan, and Chitral, identifying landslide triggering factors such as lithology, slope gradient, rainfall intensity, and seismic activity
- Executed geotechnical engineering tests including direct shear test, triaxial shear test, unconfined compression test, uniaxial compressive strength (UCS), point load test, Brazilian tensile strength test, Atterberg limits (liquid limit, plastic limit, shrinkage limit), Casagrande plasticity chart classification, and sieve analysis for grain size distribution
- Utilized advanced software tools for slope mapping and geotechnical analysis including Slide2 (Rocscience), Plaxis 2D/3D, GeoStudio (SLOPE/W, SEEP/W), RocFall, Dips, Swedge, RS2, and RockWorks for kinematic analysis, slope stability modeling, and rockfall simulation
- Conducted detailed landslide susceptibility mapping, impact analysis, and risk assessment using integrated field data and remote sensing techniques
- Prepared comprehensive geological maps, cross-sections, and structural contour diagrams for various terrains, contributing to disaster risk reduction strategies

CERTIFICATIONS

- **Remote Sensing & Geospatial:** Satellite Imagery, Remote Sensing & Machine Learning Specialization (Coursera); Spatial Data Science and Applications (Yonsei University, Coursera); Spatial Data Science: The New Frontier in Analytics (Esri MOOC); GIS for Climate Action (Esri MOOC); Imagery in Action (Esri MOOC); Hyperspectral Thermal Image Unmixing (Geo University); Active and Passive Satellite Data Analysis Using Cloud Computing for Surface Water/Flood Mapping
- **Deep Learning & AI:** IBM AI Engineering (Coursera); Machine Learning Specialization; Deep Learning Specialization; Image and Video Processing: From Mars to Hollywood with a Stop at the Hospital (Duke University); Fundamentals of Digital Image and Video Processing (Coursera)

LANGUAGES & ACTIVITIES

- **Languages:** Urdu - Native; English - (TOEFL)
- **Social and Multicultural:** Co-founder, Hamdard Welfare Foundation Azad Jammu and Kashmir; Member, Alkhidmat Foundation Pakistan; Climate Change Activist focusing on Multi-Hazards in Pakistan; Member, NUST Environment Club
- Active memmber of Blood Chain Pakistan, saving lives drop by drop through blood donations telesthesia and blood cancer patients

ACADEMIC REFERENCES

DR. JAVED IQBAL javed@igis.nust.edu.pk
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